

RESEARCH DIRECTION

- **High Performance Computing, Parallel Programming, and Scientific Computing.**

EDUCATION

- **Peking University** **Beijing, China**
Undergraduate Student *Sep. 2025-Now*
 - 2026.09 - Now, Theoretical and Applied Mechanics (Engineering and Scientific Computing track), College of Engineering.
 - 2025.09 - 2026.06, Chemistry (Materials Science and Engineering track), College of Engineering.

Mathematics & Physics: Linear Algebra and Geometry (98); Mathematical Analysis (II) (92); Mathematical Analysis (I) (91); General Physics (I) (91)

Computer Science: Introduction to Computing (B) (96); Data Structures and Algorithms (B) (94); Parallel Programming (91)

Term GPA: 3.87, Fall 2025
- **Changjun High School** **Changsha, Hunan, China**
High School Student *Sep. 2022-Jun. 2025*

SELECTED HONORS

- **Grand Prize, National Level, National High School Innovation Writing Competition (Senior 3 Group)**
National-level grand prize in the senior high school graduating class group.
- **First Prize, Provincial Level, "FLTRP Cup" Foreign Language Literacy Competition**
Provincial-level first prize in the foreign language literacy competition.
- **First Prize, Hunan Region, 2023 Chinese Mathematical Olympiad Preliminary Round (Chinese High School Mathematics League)** 2023
Provincial first prize in the Hunan division. [Link](#) [PDF](#)
- **Second Prize, Beijing Region, Chinese College Students Mathematics Competition** 2025
17th National College Students Mathematics Competition / 36th Beijing College Students Mathematics Competition. [Link](#) [PDF](#)
- **Admitted to Zhou Pei-Yuan College, Peking University** 2026
One of 15 students admitted from the 362-member 2025 College of Engineering cohort.

RESEARCH EXPERIENCE

- **Assoc. Prof. Mohan Chen's Research Group** **Beijing, China**
Undergraduate Research Training *2026.01-Now*
 - The group is affiliated with Peking University Academy for Advanced Interdisciplinary Studies.
 - Originally, the `hsolver` module in the open-source software used the CG or BPCG method. I reproduced the PPCG method from the paper, and the implementation is currently under review in [ABACUS Pull Request #7580](#) below.

PUBLICATIONS

- **Solving Functional Equations via the Injection-Commutation Method** 2023
[Link](#) · [PDF](#)
CNKI-indexed journal article, published in Middle School Mathematics, 2023(5).

RELEVANT COURSEWORK AND SELF-STUDY

- **Introduction to Embodied Intelligence**, by He Wang
Audited course on robotics-oriented AI systems.
- **Introduction to Computer Vision**, by He Wang
Audited course on visual perception foundations.
- **Introduction to Artificial Intelligence**
Audited course on AI foundations and methods.
- **Large Models: From Foundations to Practice**
Summer school course on large-model foundations and applications.
- **CS231n: Deep Learning for Computer Vision**
Self-study of deep learning for computer vision.

SELECTED TECHNICAL WORK

- **Leyun SparkEdu** 2026
Technical project experience from DAMO Developer Matrix, Alibaba DAMO Academy, with exposure to VLA (Vision-Language-Action) model training workflows.
 - Participated in a technical project involving interaction scenarios and VLA model training workflows.
 - Used the project as a focused bridge between mathematical computing and programming practice.
- **ABACUS Pull Request #7580** [Link](#) 2026
PPCG iterative diagonalization solver for an open-source electronic-structure package.
 - Proposed a PPCG iterative diagonalization solver and added the `ks_solver=ppcg` path.
 - Added unit tests, build integration, and user-facing documentation.
 - Practiced reading and modifying a large research codebase involving numerical kernels and build-system constraints.

TECHNICAL SKILLS

Programming	Python (proficient), C++, HTML
Computing	Linux, Git, CMake, MPI, OpenMP, CUDA, LAPACK-related workflows
Foundations	Linear algebra
Writing	Markdown, LaTeX-style mathematical writing